Final Project Outline

Cricket Bergner

1 Introduction

- The final project studies the chaotic motion of the magnetic pendulum/double pendulum.
- A brief explanation of how the chosen pendulum works.
- Conceptually interesting things, how they may influence the algorithm, errors that may emerge.
- Nice transition statement.

1.1 Languages and Libraries

- Code written in Python.
- List the libraries.
  - New/interesting libraries that lent themselves to new skills or performed new tasks.

2 Undergirding Theory

- The Magnetic/Double Pendulum.

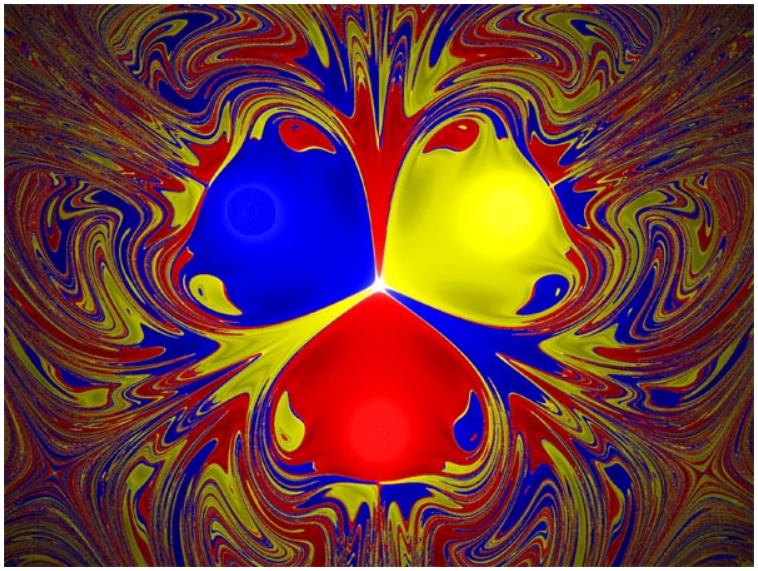

Figure 1: Placeholder image.

- Chaotic motion and what interesting numerical results it brings forward.
- Some explanation on fractals, and what makes them interesting mathematically.
- The chosen symplectic integrator (and relevant equations).

3 Walking Through the Code

- Main Body (3 Magnet Magnetic Pendulum)
  - Map of motion over time.

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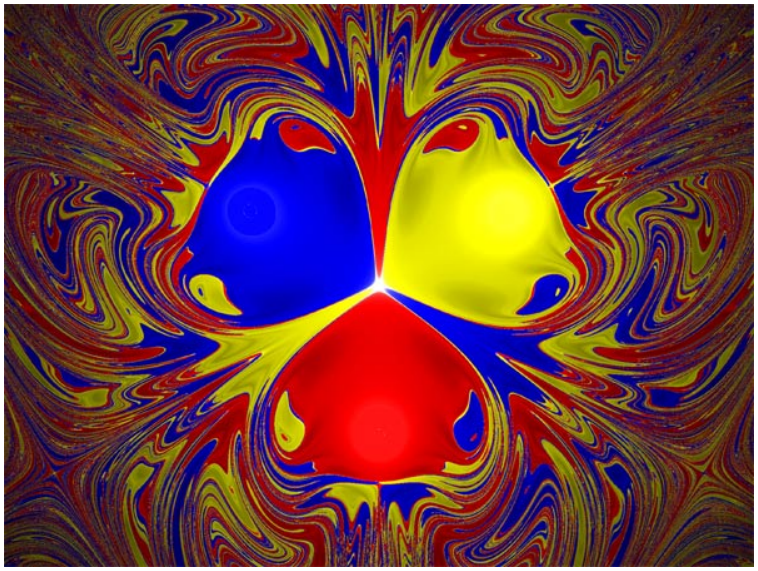

Figure 2: Placeholder image.

- Map of the fractal it creates.

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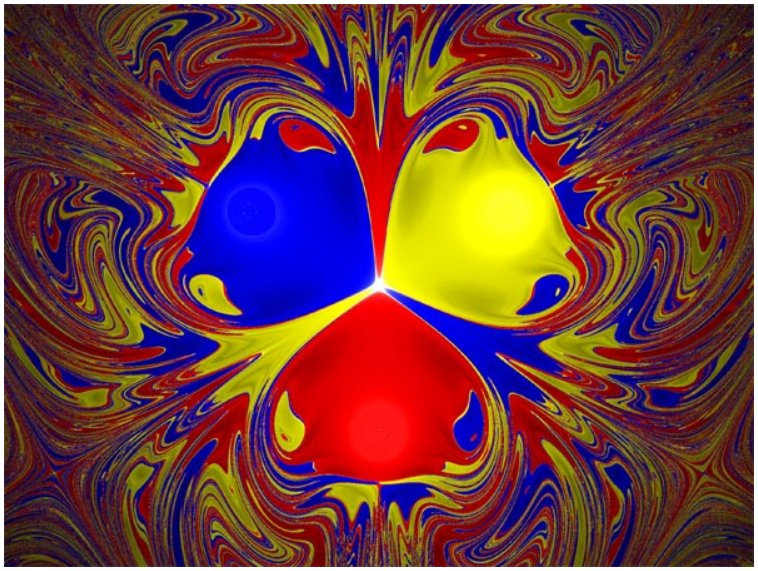

Figure 3: Placeholder image.

- Extension 1 (4 Magnet Magnetic Pendulum)
  - Map of motion over time.

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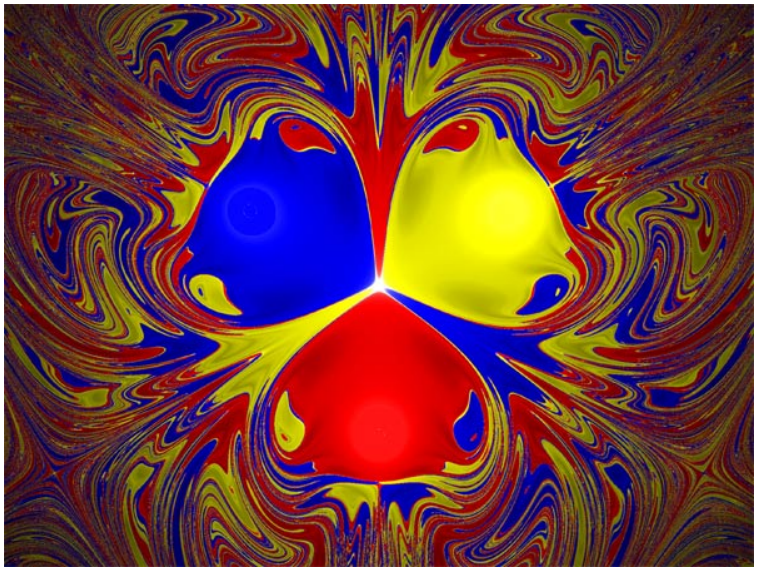

Figure 4: Placeholder image.

- Map of the fractal it creates.

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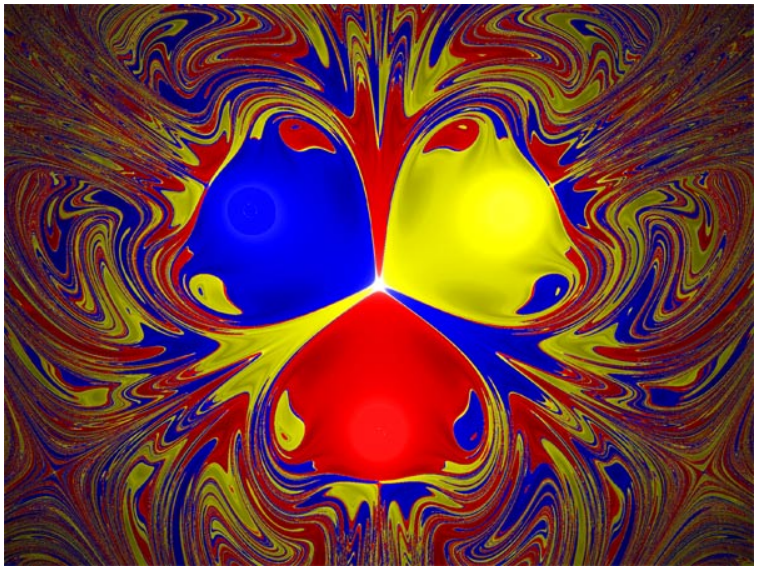

Figure 5: Placeholder image.

- Extension 2 (5 Magnet Magnetic Pendulum)
  - Map of motion over time.

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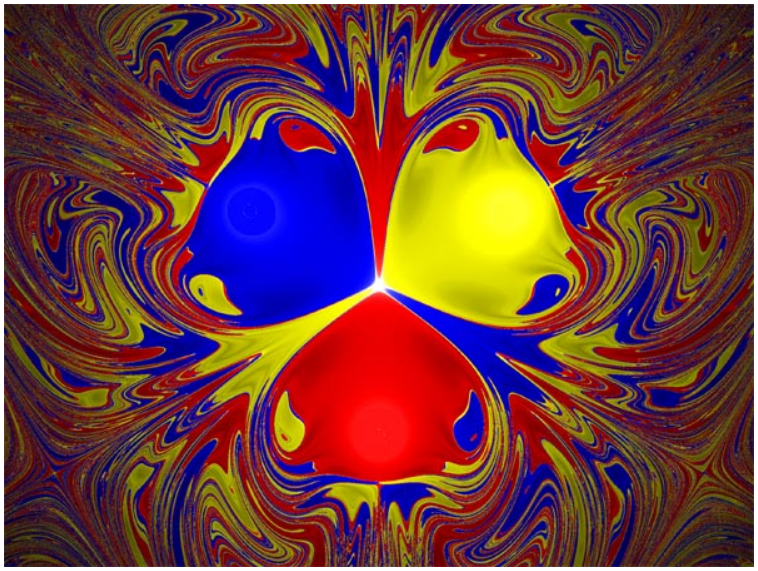

Figure 6: Placeholder image.

- Map of the fractal it creates.

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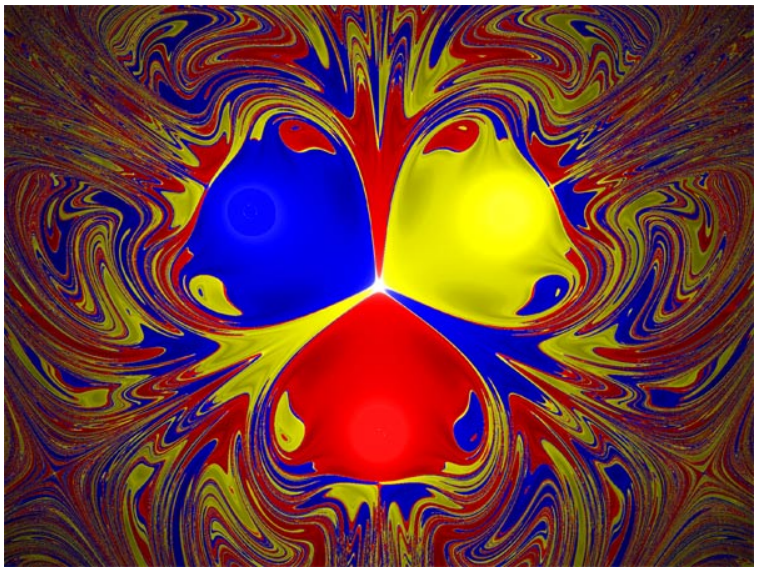

Figure 7: Placeholder image.

4 Lessons Learned

- Interesting results.
- New organization or calculation techniques.
- Additional complications and my takeaways from them.

4.1 Time Keeping

- How much time on the assignment.
- How much time on the writeup.
- How much time on research/asking questions.
- How much time on slides.

4.2 Attributions

- Important links/articles/books/people.